Simulation-Based Inference

Introduction to Simulations

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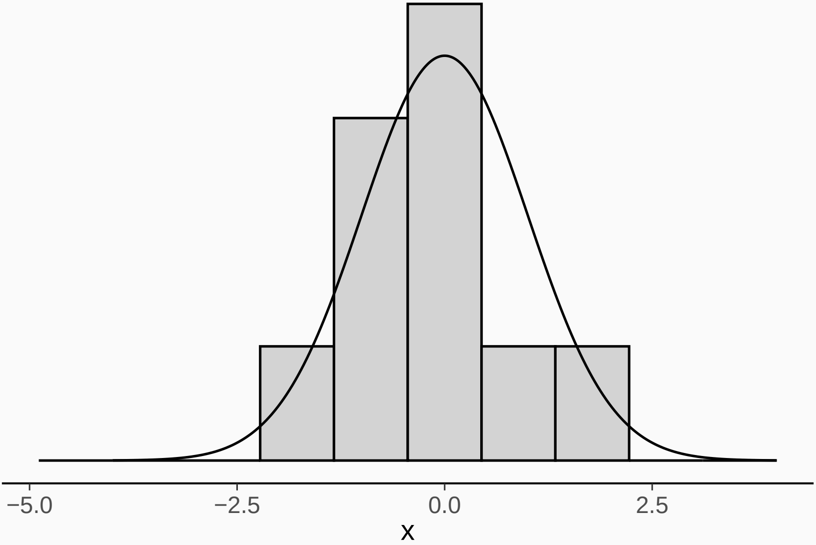
“Random” Sampling (Simulations)

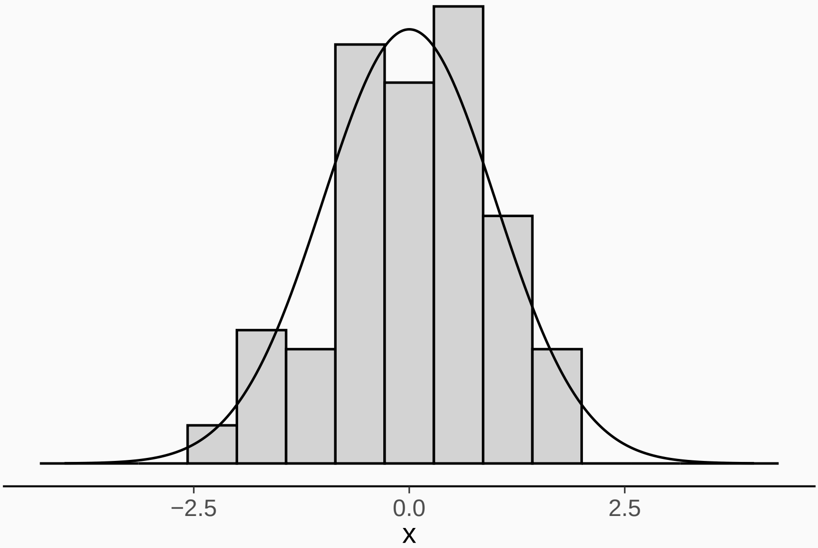
Obtain S values $x^{(s)}$ $(1 \leq s \leq S)$ of a variable x in a way that their histogram matches the density of the targeted distribution infinitely well for infinite number of draws.

- The simulated $x^{(s)}$ values are called *draws* or *samples*
- Computers are deterministic and can only obtain “pseudo-random” draws using pseudo-random number generators
- Simulations can be reproduced by setting the same random *seed*

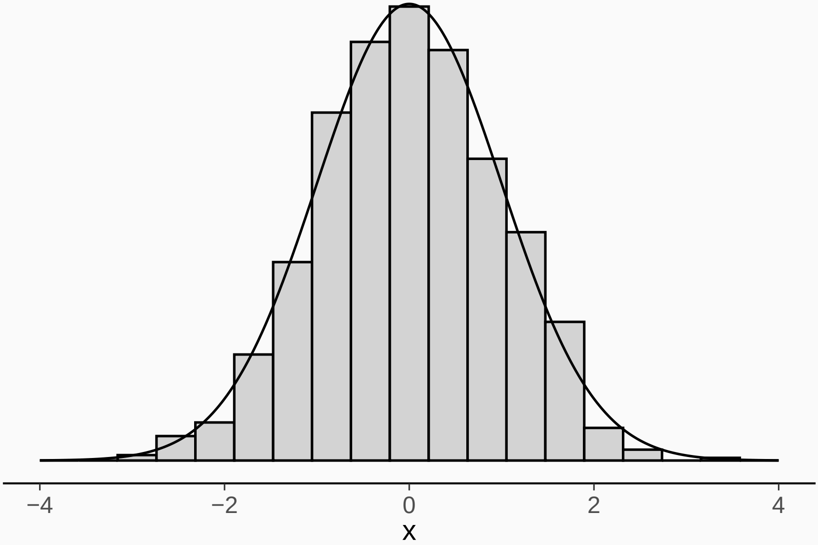
Sampling: Univariate Visualization (10 draws)



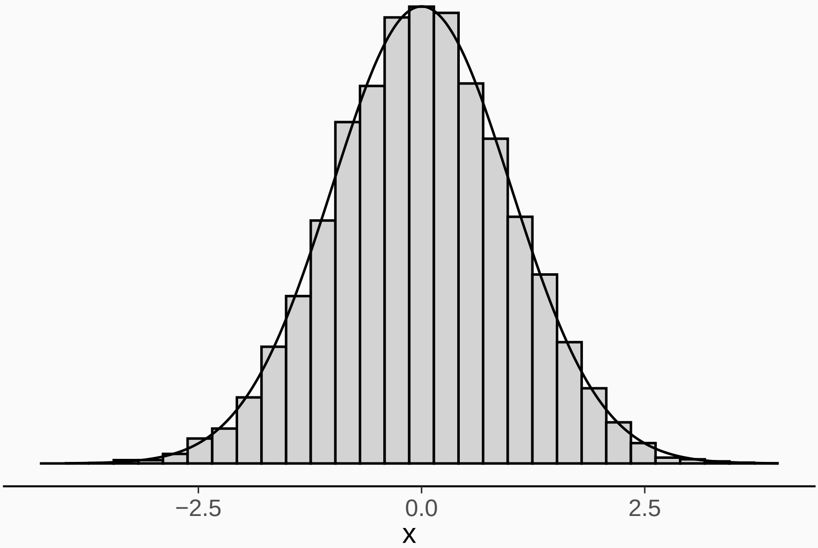
Sampling: Univariate Visualization (100 draws)



Sampling: Univariate Visualization (1000 draws)



Sampling: Univariate Visualization (10000 draws)



Using Draws to Approximate Expectations

(Almost) every quantity of interest related to x is an expectation over $p(x)$:

$$\mathbb{E}_x(f(x)) = \int f(x)\, p(x)\; \mathrm{d}x$$

Having obtained independent random draws $\{x^{(s)}\}$ from $p(x)$:

$$\frac{1}{S} \sum_{s=1}^{S} f(x^{(s)}) \sim \text{Normal}\left(\mathbb{E}_x(f(x)), \sqrt{\frac{\text{Var}_x(f(x))}{\text{S}}} \right)$$

The above standard deviation is called Monte Carlo Standard Error (MCSE)

$$\text{MCSE} = \sqrt{\frac{\text{Var}_x(f(x))}{\text{S}}}$$

Suppose we have the density $p_x(x)$ of x

What is the density $p_y(y)$ of $y = \exp(x)$?

Jacobian adjustment

Assume $x \in \mathbb{R}$ and $f : \mathbb{R} \to \mathbb{R}, x = f(y)$ to be continuously differentiable, then

$$\int_{\Omega_x} p_x(x) dx = \int_{\Omega_y} p_x(f(y)) \, |f'(y)| \, dy$$

where $f'(y)$ is the first derivative of f wrt. y

Also known as "integration by substitution"

Jacobian adjustment

Since $\int_{\Omega_x} p_x(x)dx = 1$, we have

$$p_y(y) = p_x(f(y))\,|f'(y)|$$

Usually, we start from $y = g(x)$ and compute $f = g^{-1} : y \to x$

In higher dimensions, $f'(y)$ is replaced by the determinant of the Jacobian matrix $J_f(y)$ of f wrt. y

Jacobian adjustment: lognormal example

Suppose that x is standard normally distributed

$$p_x(x) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2} x^2)$$

Consider $y = g(x) = \exp(x)$ such that $x = f(y) = \log(y)$

Accordingly, $\frac{dx}{dy} = f'(y) = \frac{1}{y}$ such that $dx = \frac{1}{y} dy$

We obtain the density of the standard lognormal distribution:

$$p_y(y) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2} \log(y)^2) \ \frac{1}{y}$$

Change of variables via random draws

If we have sampled S draws $x^{(s)}$ from $p(x)$

$$x^{(s)} \sim p(x)$$

then the transformed draws $f(x^{(s)})$ are from $p(f(x))$

$$f(x^{(s)}) \sim p(f(x))$$

No Jacobian adjustments needed!

Example: Change of variables via random draws